12 A survey conducted in 2021 showed that 10% of British adults were vegetarians.

A dietitian believes that the proportion of British adults who are vegetarians may have changed, so decides to conduct a hypothesis test at the 5% level of significance.

In a random sample of 112 adults, the dietitian finds that there are 19 vegetarians.

Carry out the hypothesis test to determine whether there is any evidence to support the dietitian's belief. **[7]**